Quiz: Investment Appraisal - Version A

FAST Lahore – Machine Learning & AI Research Block

Name: ______________________________ **Roll No:** ______________________________

Scenario:

FAST National University, Lahore Campus, is planning to construct a new dedicated **Machine Learning & AI Research Block**. The administration has compiled the following cost and revenue projections:

Initial Investment (Year 0):

Cost Component	Amount (PKR)
Land Acquisition & Site Preparation	45,000,000
Construction & Civil Works	120,000,000
High-Performance Computing Lab (GPU Clusters, Servers)	65,000,000
Smart Classroom Technology & Furniture	25,000,000
Networking Infrastructure & Security Systems	15,000,000
Total Initial Investment	**270,000,000**

Projected Annual Cash Inflows (PKR):

The block will generate revenue through increased student enrollment, executive training programs, industry-sponsored research, and government grants.

Year	Tuition & Fees	Executive Programs	Research Grants	Industry Partners	Total Inflow
1	35,000,000	12,000,000	8,000,000	5,000,000	60,000,000
2	48,000,000	18,000,000	15,000,000	12,000,000	93,000,000
3	55,000,000	24,000,000	22,000,000	19,000,000	120,000,000
4	58,000,000	28,000,000	28,000,000	26,000,000	140,000,000
5	60,000,000	30,000,000	32,000,000	28,000,000	150,000,000
6	62,000,000	32,000,000	35,000,000	31,000,000	160,000,000

Annual Operating Costs:

Note: Operating costs are incurred every year from Year 1 through Year 6 (including the final year when equipment is sold).

Cost Item	Annual Amount (PKR)
Faculty & Staff Salaries	22,000,000
Utilities & Maintenance	6,000,000
Software Licenses & Cloud Computing	8,000,000
Administrative Overhead	4,000,000
Total Annual Operating Costs	**40,000,000**

Additional Information:

- Discount rate: **14%**
- At the end of Year 6, the computing equipment will have a resale value of **PKR 18,000,000**
- Assume all cash flows occur at year-end

Required:

Part A (10 marks): Calculate the **Net Cash Flows** for each year (Year 0 through Year 6), including the resale value in the final year. Present your answer in a table.

Solution A: Net Cash Flows *( ) indicate negative value*

Year	Initial Investment	Revenue	Operating Cost	Resale	Net Cash Flow
0	(270,000,000)	-	-	-	(270,000,000)
1	-	60,000,000	(40,000,000)	-	20,000,000
2	-	93,000,000	(40,000,000)	-	53,000,000
3	-	120,000,000	(40,000,000)	-	80,000,000
4	-	140,000,000	(40,000,000)	-	100,000,000
5	-	150,000,000	(40,000,000)	-	110,000,000
6	-	160,000,000	(40,000,000)	18,000,000	138,000,000

Part B (20 marks): Using the Discounted Cash Flow (DCF) method, compute the **Present Value** of each year's net cash flow using the 14% discount rate. Present your calculations in a table showing: Year, Net Cash Flow, Discount Factor (formula: $1 \div (1+r)^t$), and Present Value.

Solution B: Present Value Calculation

Year	Net Cash Flow	Discount Formula (14%)	Present Value
0	(270,000,000)	-	(270,000,000)
1	20,000,000	$1 / (1 + 0.14)^1 = 0.8771929825$	17,543,859.65
2	53,000,000	$1 / (1 + 0.14)^2 = 0.7694675285$	40,781,779.01
3	80,000,000	$1 / (1 + 0.14)^3 = 0.6749715162$	53,997,721.30
4	100,000,000	$1 / (1 + 0.14)^4 = 0.5920802774$	59,208,027.74
5	110,000,000	$1 / (1 + 0.14)^5 = 0.5193686644$	57,130,553.08
6	138,000,000	$1 / (1 + 0.14)^6 = 0.4555865477$	62,870,943.58

Part C (5 marks): Calculate the **Payback Period** using cumulative net cash flows. Express your answer in years and months.

Solution C: Payback Period

Cumulative Cash Flows:
Year 0: (270,000,000)
Year 1: (250,000,000)
Year 2: (197,000,000)
Year 3: (117,000,000)
Year 4: (17,000,000)
Year 5: 93,000,000

Payback occurs in Year 5.
Fraction of Year 5 = 17,000,000 / 110,000,000 = 0.1545 years
Months = 0.1545 × 12 = 1.85 months (approx. 2 months)
Payback Period = 4 years and 2 months (ie 49.85 months)

Part D (10 marks): Calculate the **Net Present Value (NPV)** of the project. Based on your calculation, should FAST Lahore proceed with the construction? Justify your recommendation using the NPV decision rule.

Solution D: Net Present Value (NPV)

Sum of Present Values (Years 1 to 6) = 17,543,859.65 + 40,781,779.01 + 53,997,721.30 + 59,208,027.74 + 57,130,553.08 + 62,870,943.58 = 291,532,884.36
NPV = Total PV of Cash Inflows - Initial Investment
NPV = 291,532,884.36 - 270,000,000 = **PKR 21,532,884.36** (or if we ignore values after decimals then 21,532,882). Any value in this range is acceptable (different rouding ranges).

Recommendation: Since the NPV is positive (PKR 21,532,882 > 0), FAST Lahore should proceed with the construction of the Machine Learning & AI Research Block.

Total Marks: 45